

Table 1: Collision cost, cost_limit=0: SafeDreamer (200k steps) vs MACPO (5M steps).

Algorithm	Scenario	Score $\uparrow$	Cost $\downarrow$	Winrate $\uparrow$	Steps	Cost Limit
MACPO	2s_vs_1sc	20.16 $\pm$ 0.09	0.00 $\pm$ 0.00	0.99 $\pm$ 0.01	5M	0
Safe Dreamers	2s_vs_1sc	20.10 $\pm$ 0.21	0.00 $\pm$ 0.00	0.98 $\pm$ 0.02	200k	0
MACPO	3m	19.85 $\pm$ 0.11	0.00 $\pm$ 0.00	0.99 $\pm$ 0.01	5M	0
Safe Dreamers	3m	19.83 $\pm$ 0.04	0.02 $\pm$ 0.01	0.99 $\pm$ 0.00	200k	0
MACPO	2s3z	18.76 $\pm$ 0.05	0.03 $\pm$ 0.02	0.80 $\pm$ 0.01	5M	0
Safe Dreamers	2s3z	18.96 $\pm$ 0.41	0.12 $\pm$ 0.02	0.82 $\pm$ 0.07	200k	0
MACPO	3s_vs_3z	19.55 $\pm$ 0.35	0.00 $\pm$ 0.00	0.94 $\pm$ 0.05	5M	0
Safe Dreamers	3s_vs_3z	19.73 $\pm$ 0.31	0.06 $\pm$ 0.01	0.96 $\pm$ 0.03	200k	0
MACPO	3s_vs_4z	9.18 $\pm$ 0.86	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
Safe Dreamers	3s_vs_4z	18.80 $\pm$ 1.57	0.20 $\pm$ 0.15	0.50 $\pm$ 0.26	200k	0
MACPO	3s_vs_5z	7.16 $\pm$ 0.76	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	5M	0
Safe Dreamers	3s_vs_5z	18.91 $\pm$ 1.64	0.04 $\pm$ 0.04	0.40 $\pm$ 0.14	200k	0
MACPO	8m	19.31 $\pm$ 0.79	0.34 $\pm$ 0.05	0.92 $\pm$ 0.10	5M	0
Safe Dreamers	8m	16.67 $\pm$ 0.51	0.43 $\pm$ 0.12	0.62 $\pm$ 0.05	200k	0
MACPO	MMM	11.13 $\pm$ 0.17	0.96 $\pm$ 0.10	0.00 $\pm$ 0.00	5M	0
Safe Dreamers	MMM	17.34 $\pm$ 1.01	0.55 $\pm$ 0.29	0.65 $\pm$ 0.07	200k	0
MACPO	3s5z_vs_3s6z	9.56 $\pm$ 0.04	0.07 $\pm$ 0.03	0.00 $\pm$ 0.00	5M	0
Safe Dreamers	3s5z_vs_3s6z	11.87 $\pm$ 1.23	0.83 $\pm$ 0.36	0.06 $\pm$ 0.10	200k	0
MACPO	bane_vs_bane	—	—	—	5M	0
Safe Dreamers	bane_vs_bane	18.19 $\pm$ 0.38	0.17 $\pm$ 0.24	0.57 $\pm$ 0.06	200k	0